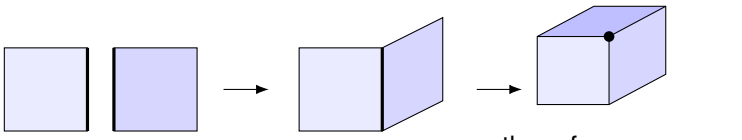
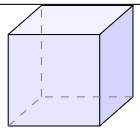
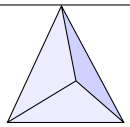
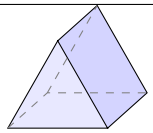
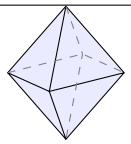
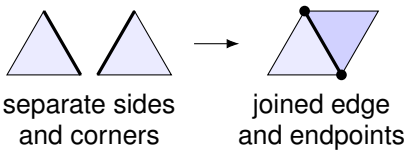
Use closed models with all their faces attached. A vertex is a corner of the assembled solid; an edge is shared by two faces. Count each assembled edge and vertex once. Tabs and glue overlaps do not count.




**Problem 1:** Find the vertices, edges and faces of each closed model.

| Model | Vertices $V$ | Edges $E$ | Faces $F$ |
|---|---|---|---|
|  cube | | | |
|  tetrahedron | | | |
|  triangular prism | | | |
|  octahedron | | | |

**Problem 2:** Imagine cutting every face free. Find a rule that turns the number of separate face sides into the assembled edge count. Can the separate face-corner count alone tell you the vertex count?

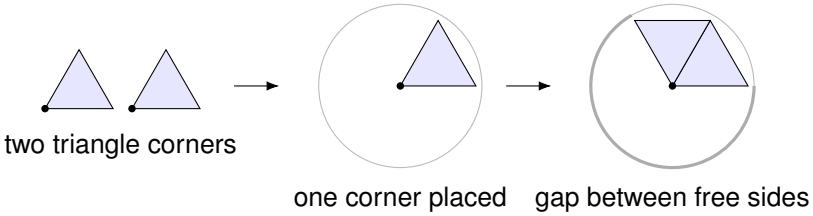


separate sides and corners

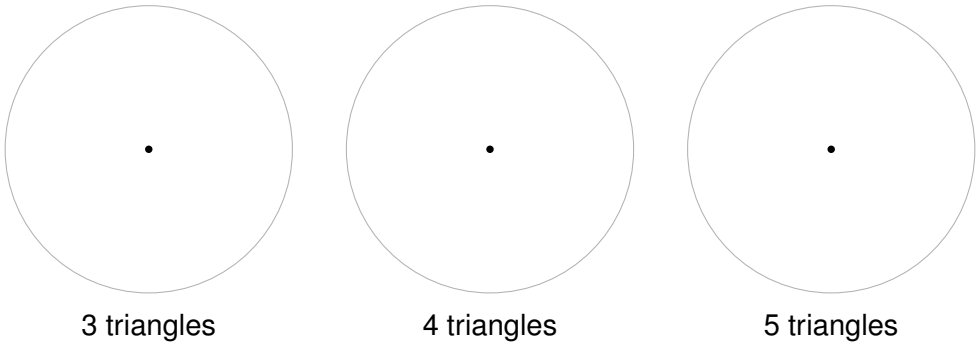
joined edge and endpoints

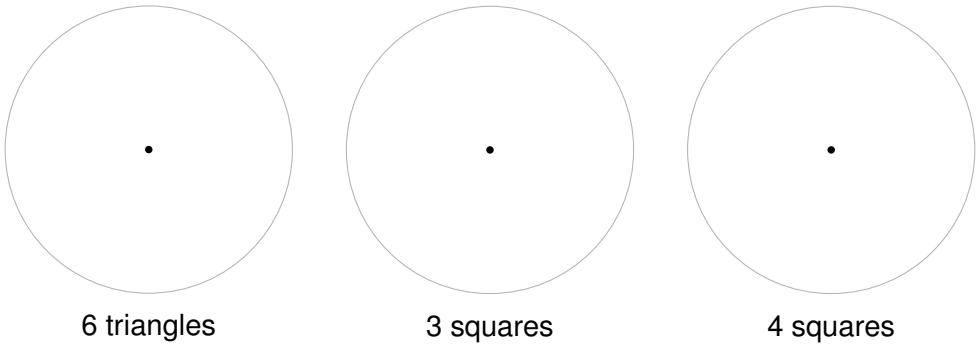
| Faces before assembly | Separate sides | Assembled edges | Separate corners | Assembled vertices |
|---|---|---|---|---|
| 6 squares: cube | | | | |
| 4 triangles: tetrahedron | | | | |
| 2 triangles and 3 squares: prism | | | | |
| 8 triangles: octahedron | | | | |

Join face corners around one point. Keep their sides meeting edge to edge and their marked vertices together. The gap is the missing part of a full turn when the corners lie flat.

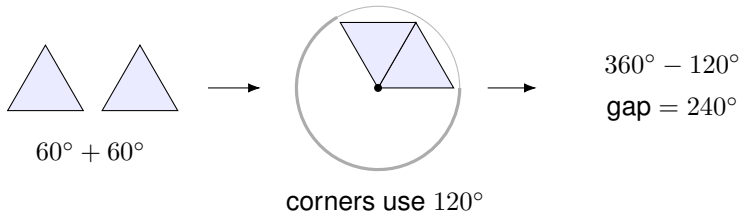



**Problem 3:** Which groups can close into a pointed corner with no inward dents? Which close flat with no gap? Make each group, then draw its gap with the corners laid flat.



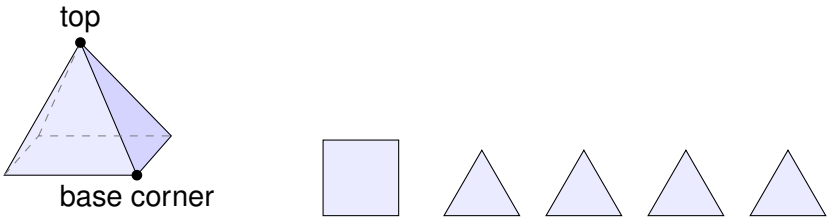
A full turn is $360°$. An equilateral triangle corner is $60°$; a square corner is $90°$. A vertex's gap is $360°$ minus its face-corner angles.




**Problem 4:** Find the gap at every vertex of each closed model. How many full turns do all its gaps make together?

| Model | Vertices | Face corners at one vertex | Gap at one vertex | All gaps together |
|---|---|---|---|---|
| Cube | | | | |
| Tetrahedron | | | | |
| Triangular prism | | | | |
| Octahedron | | | | |

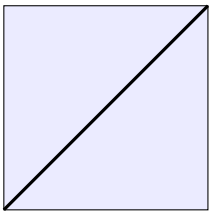
**Problem 5:** The square pyramid has one square face and four equilateral triangle faces. Which corner leaves the bigger gap, the top or a base corner? Find the pyramid's total gap.



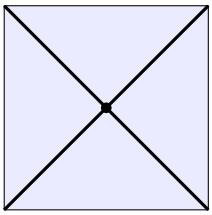

| Vertex type | How many? | Face corners meeting there | Gap at each one |
|---|---|---|---|
| Top | | | |
| Base corner | | | |

In a redrawn surface, marked points count as vertices, segments between vertices count as edges, and each face piece counts as a face.

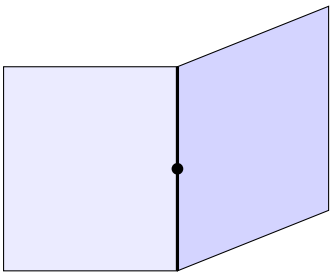
**Problem 6:** Compare $V - E + F$ for all four models in Problem 1. Keep the cube's shape and try each of these three changes on its own, starting from the original cube. What happens to $V - E + F$ and to the total gap? Can a new vertex have a gap while the cube keeps its shape?



one face divided by a diagonal



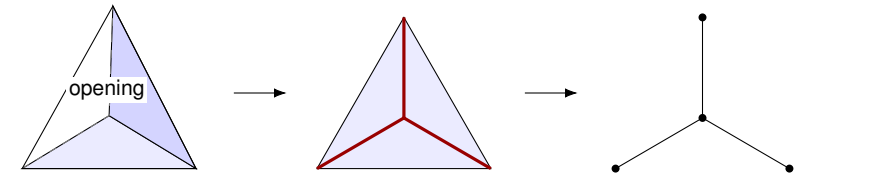
one face divided from its center



one vertex inserted along a shared edge

| Closed surface | $V$ | $E$ | $F$ | $V - E + F$ | Total gap |
|---|---|---|---|---|---|
| Original cube | | | | | |
| One diagonal | | | | | |
| Center and 4 lines | | | | | |
| One edge vertex | | | | | |

Convex means no dents. An opened drawing preserves the assembled edge and vertex connections. Count its bounded regions as faces; the outside is the removed face. A tree is a connected drawing with no closed loops.

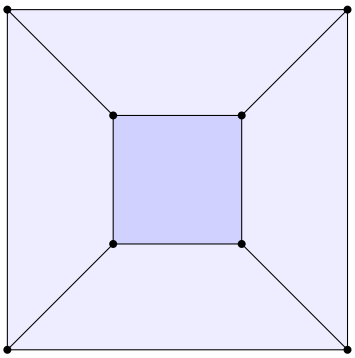



tetrahedron with one face removed

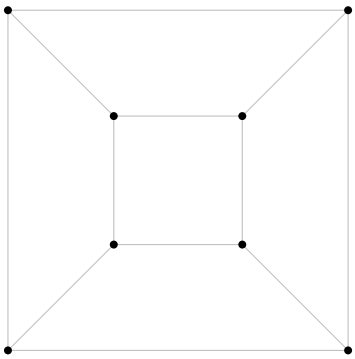
same three faces viewed through the opening

three boundary edges erased a tree remains

**Problem 7:** Explain why $V - E + F$ always has the same value for closed convex solids with polygon faces. Can an opened drawing be changed into a tree while keeping all its vertices connected?

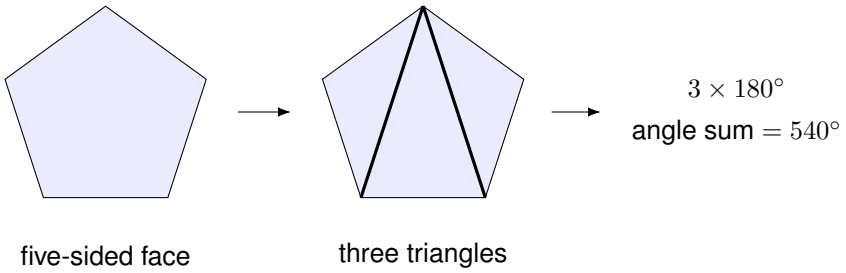


opened cube: five bounded faces



extra workspace

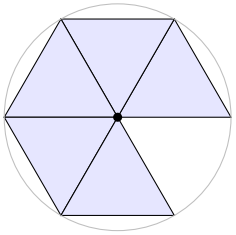
The angles of a triangle add to $180°$. A polygon with $n$ sides can be divided into $n - 2$ triangles, so its corner angles add to $180(n - 2)°$.



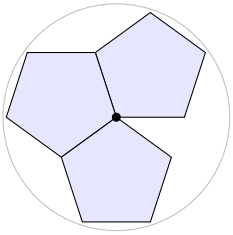

**Problem 8:** Explain why every closed convex solid with polygon faces has the same total gap, even when its vertex gaps are different.

A regular polygon has equal sides and equal angles. Every face of each solid below is one regular polygon.

**Problem 9:** Find the number of vertices, edges and faces each closed convex solid with polygon faces must have, without a model of the whole solid.



Triangle faces; five meet at every vertex



Pentagon faces; three meet at every vertex

| Solid | Vertices | Edges | Faces |
|---|---|---|---|
| Five triangles at each vertex | | | |
| Three pentagons at each vertex | | | |